Ranking, Calibration and Cost in Competing-Risks Survival Models

A Common-Protocol Benchmark on 550,000 Mortgages

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The question

Deep competing-risks models are selected and reported on **ranking**. Lenders act on the **probability**. Does the advantage survive when the probabilities are measured too?

Why it matters

A mortgage ends by default, prepayment or maturity — once one happens the others cannot. Provisioning needs the chance of default *by each date*, not an ordering of borrowers.

A model can rank borrowers perfectly and still attach the wrong numbers to them. Reserves are computed from the numbers.

Data

Freddie Mac Single-Family Loan-Level Dataset, 50,000-loan samples per vintage.

Cohort	Train	Test
Post-crisis	2013–15	2016–18
Crisis	2005–06	2007–09

20.6% vs 95.5%

censoring, counting prepayment as an event versus censoring it

Dirick et al. report 20–85% and 94.56–98.16% on European consumer credit. The same effect, on a different product, country and decade.

Protocol

One interface: every model returns a cumulative incidence per cause, so none can be given a metric the others cannot.

Models Aalen–Johansen (null), cause-specific Cox, Fine–Gray, random survival forest, DeepHit

Ranking IPCW time-dependent AUC

Calibration integrated calibration index, integrated Brier

Cost fit and predict seconds

Splits vintage-ordered, not random

Budget 20 DeepHit configs, 8 forest, fixed in advance, validated out-of-time

Inference paired bootstrap, $B{=}500$, two-sided, α fixed at 0.05, Bonferroni across three metric families

1. The advantage reverses with regime

AUC at 60 months, default:

	Crisis	Post-crisis
DeepHit (tuned)	**0.8538**	0.7280
Cause-specific Cox	0.8346	**0.7372**
Difference	$+0.0192$	-0.0091

Both significant. But the Cox model's own training variability on crisis data is ± 0.0159, so the win is barely one standard deviation of the baseline's noise — the *classical* models are the unstable ones here.

The calibration penalty does **not** reverse: $+0.0040$ worse than Cox in both cohorts.

2. No gain over a random forest

$p = 0.97$ $p = 0.79$

DeepHit vs tuned random survival forest, AUC, crisis and post-crisis

Both differences fall inside DeepHit's own seed variation. On crisis **prepayment** the forest is clearly better: AUC 0.685 vs 0.620, calibration index 0.154 vs 0.236.

3. The loss weight is not a reliable knob

DeepHit's loss is α·likelihood + $(1-\alpha)$·ranking. **Pooled** over our 40 tuning configurations, validation AUC climbs $0.658 \rightarrow 0.776$ as α goes $0.1 \rightarrow 1.0$ — a clean monotone story.

It does not survive control. Everything else held **fixed**, five seeds, test vintages:

Cause	Cohort	ΔAUC	seed sd
Default	crisis	$+0.0075$	0.0033
Default	post	$+0.0005$	0.0018
Prepay	crisis	$\mathbf{-0.0673}$	0.0103
Prepay	post	$\mathbf{+0.0386}$	0.0062

Δ is $\alpha{=}0.1 \rightarrow \alpha{=}1$. The effect **reverses sign** across causes. There is no α we can recommend.

A marginal taken over a random search is not an ablation. In the search α varied alongside everything else. We had written the opposite conclusion before we ran the controlled version.

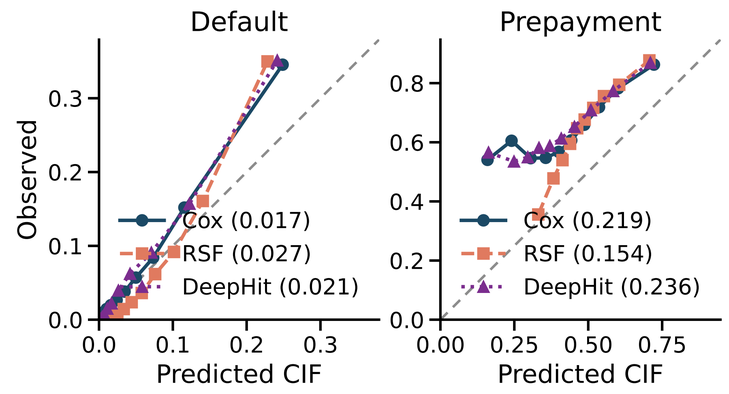
4. Recalibration cannot fix it

Calibration index	Crisis	Post-crisis
Raw	0.0209	0.0318
Recalibrated (train)	0.0164	0.0297
Recalibrated (oracle)	**0.0053**	**0.0082**

Fitted on the training vintages — all a lender has — rescaling changes almost nothing, and in 10 of 16 cells makes calibration worse. Fitted on the test vintages, an oracle nobody can build, it cuts error by a median factor of 4.2.

So the penalty is not a broken calibration map. It is the vintage shift itself: correctable in principle, unavailable in practice.

Calibration at 60 months, crisis cohort



Deciles of predicted risk against the Aalen–Johansen estimate. Trained on 2005–06, tested on 2007–09, *every* curve sits above the diagonal — prepayment worst.

Cost

$\approx 2{,}000\times$

DeepHit (59 s) versus Fine–Gray (0.03 s) per fit, 150,000 loans

Plus 14.4 minutes of hyperparameter search the classical models never need.

What we cannot claim

One dataset family. Only five bootstrap replicates per model. Covariates observed at origination only — no time-varying macroeconomic variables. Our recalibration is a marginal rescaling, not the per-cause procedures of Alberge et al. (2026). With 150,000 test loans almost anything is significant, so we report effect sizes against training noise.

Better ranking in one regime.

Worse probabilities in all of them.

Code: `github.com/ar-sadraei/rtds-project`

Data: Freddie Mac SFLLD, free on registration